

EE69210: Machine Learning for Signal Processing Laboratory
Department of Electrical Engineering, Indian Institute of Technology, Kharagpur

Speech Gender Recognition using Deep Neural Networks

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Keywords: Fully connected networks, Speech Classification

Grading Rubric

	Tick the best applicable per row			Points
	Below Ex- pectations	Lacking in Some	Meets all Ex- pectations	
Completeness of the report				
Organization of the report (5 pts)				
Quality of figures (5 pts)				
Creation of Dataset and explanation of the preprocessing steps				
Model training, inference, and ex- pla- nation for the base model with discus- sions. (25 pts)				
Generation of the confusion matrix and discussion. (25)				
Discussion on how to improve accu- racy. (5 pts)				
TOTAL (100 pts)				

1. Training Details

1.1 Model Architecture

A fully connected network defined as $\text{net}(\cdot)$ is used in this study. The $\text{net}(\cdot)$ maps $\mathbb{R}^{168 \times 1} \rightarrow \mathbb{R}^{2 \times 1}$ and is defined as follows:

- (1) **Fully Connected:** Takes the input and outputs to 256 neurons.
Dropout: Probability of 0.3
- (2) **Fully Connected:** Fully connected layer with 256 output neurons.
Non-Linearity: The layer is followed by a ReLU activation.
Dropout: Probability of 0.3
- (3) **Fully Connected:** Fully connected layer with 128 output neurons.
Non-Linearity: The layer is followed by a ReLU activation.
Dropout: Probability of 0.3
- (4) **Fully Connected:** Fully connected layer with 128 output neurons.
Non-Linearity: The layer is followed by a ReLU activation.
Dropout: Probability of 0.3
- (5) **Fully Connected:** Fully connected layer with 64 output neurons.
Non-Linearity: The layer is followed by a ReLU activation.
Dropout: Probability of 0.3
- (6) **Output Layer:** Final layer of the network, with one output neuron for each class in the dataset.
Non-Linearity: Sigmoid activation is used.

1.2 Classification Loss Function

The classification task in this study uses the cross-entropy loss function, which is defined for a single sample as:

$$\mathcal{L}_{\text{CE}} = - \sum_{i=1}^C y_i \log(\hat{y}_i)$$

where C is the number of classes, y_i is the ground truth label (one-hot encoded), and $\hat{y}_i$ is the predicted probability for class i (usually obtained after applying the softmax function). The loss encourages the predicted distribution $\hat{y}$ to be close to the true distribution y .

1.3 Adam Optimizer

Adam (Adaptive Moment Estimation) is an optimization algorithm that adjusts the learning rate for each parameter using estimates of first and second moments of the gradients. The parameter update rule at each time step t is:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t} \\ \theta_t &= \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \end{aligned}$$

where g_t is the gradient at time t , m_t and v_t are the moving averages of the gradients and squared gradients, β_1 and β_2 are hyperparameters, and η is the learning rate. ε is a small constant to avoid division by zero.

2. Feature Extraction

Mel Spectrogram:

$$\text{Mel Spectrogram} = \text{TriangularFilterbank} \times \text{STFT} \quad (1)$$

The Mel spectrogram represents the spectrum of a signal, where the frequency axis is converted to the Mel scale. It is obtained by dividing the audio signal into short time frames using the Short-Time Fourier Transform (STFT), and then transforming each frame's spectrum into the Mel scale using a filter bank of triangular filters.

MFCC (Mel-Frequency Cepstral Coefficients):

$$\text{MFCC} = \text{DCT}(\log(\text{Mel Spectrogram})) \quad (2)$$

MFCC is a feature extraction technique commonly used in speech and audio signal processing. It involves computing the Mel spectrogram, taking the logarithm of the spectrogram to approximate the human ear's response, and applying the Discrete Cosine Transform (DCT) to obtain a set of coefficients. MFCCs effectively capture essential features of audio signals and are widely used in tasks such as speech recognition, speaker identification, and audio classification.

3. Dataset Details

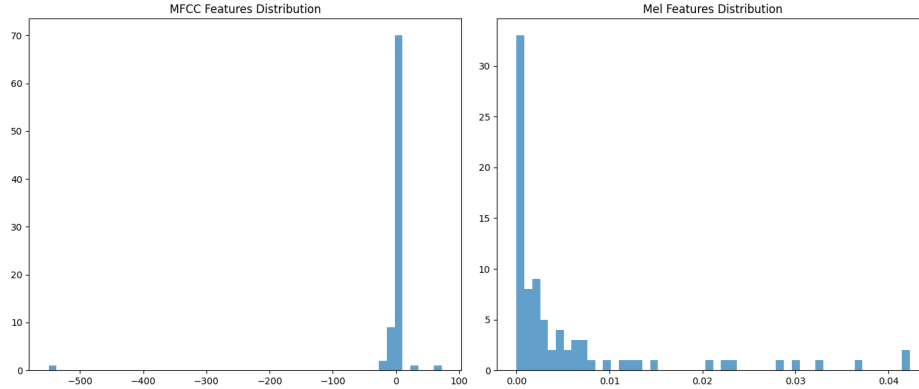


Fig. 1. Histogram corresponding to MFCC and Mel features for a given batch of data

Table 1. **Dataset Statistics**

Category	Class 0 (Male)	Class 1 (Female)
Train Set (61,068 samples)	45,840	15,228
Validation Set (7,634 samples)	5,730	1,904
Test Set (7,634 samples)	5,731	1,903

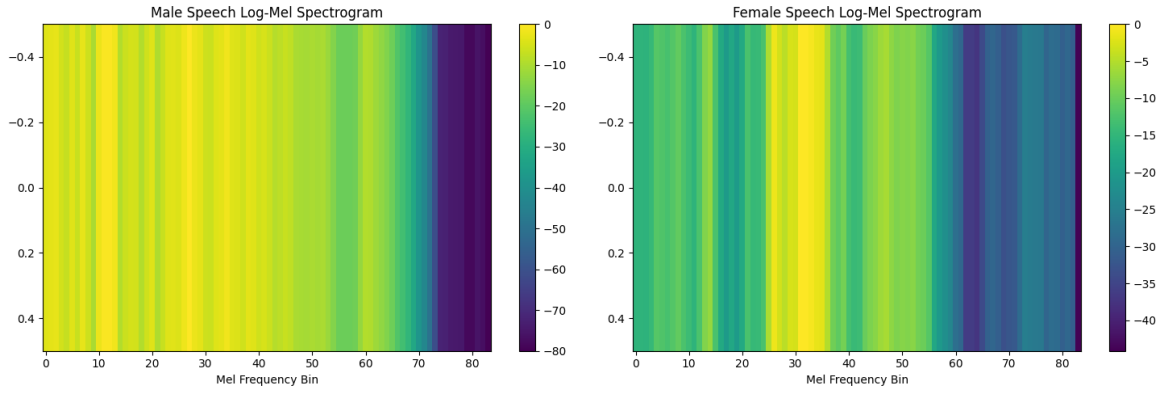


Fig. 2. Log-mel Spectrogram corresponding to male and female voices

4. Observations and Results

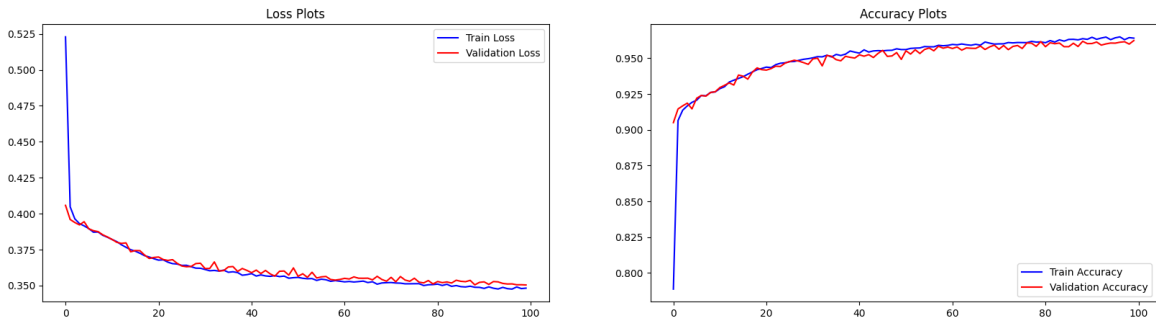


Fig. 3. Accuracy and loss plots vs training epochs

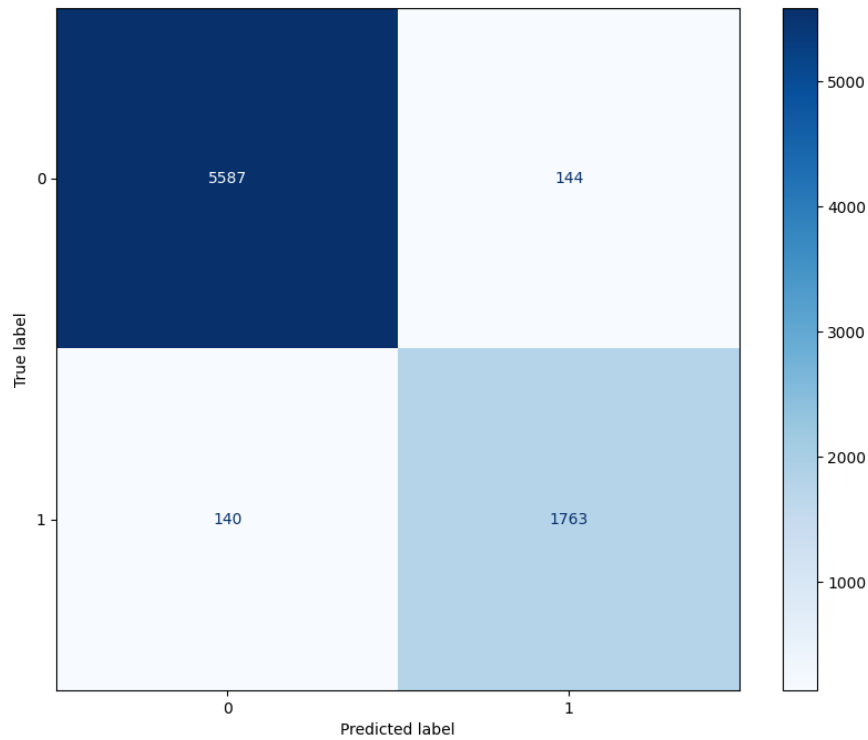


Fig. 4. Confusion matrix after prediction on test set

Classification Report:					
	precision	recall	f1-score	support	
0	0.98	0.97	0.98	5731	
1	0.92	0.93	0.93	1903	
accuracy			0.96	7634	
macro avg	0.95	0.95	0.95	7634	
weighted avg	0.96	0.96	0.96	7634	

Fig. 5. Classification report after prediction on test set

5. Discussion

5.1 Observation Discussion

The fully connected neural network trained in this study demonstrates a strong ability to classify gender based on speech features. The use of both Mel spectrograms and MFCCs provided robust feature representations, effectively capturing the spectral and perceptual characteristics of speech signals. As evidenced in the loss and accuracy plots, the model converges well over training epochs, indicating that the learning process was stable and effective.

The confusion matrix shows that the model is able to distinguish between male and female voices with high accuracy. However, a small number of misclassifications can still be observed, suggesting that some samples contain overlapping features or noisy representations that make them harder to classify correctly.

The classification report further supports the quantitative performance, showing high precision, recall, and F1-scores for both classes. This indicates that the model performs consistently across both gender categories and does not exhibit strong class imbalance issues in prediction.

5.2 Further Improvements

Although the model achieved good performance, several enhancements could be explored to improve its generalization and robustness:

- **Data Augmentation:** Introducing techniques like time stretching, pitch shifting, or adding background noise can help the model learn invariant features and improve performance on real-world, noisy data.
- **Alternative Architectures:** Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are better suited for processing spectrograms and sequential audio features, respectively. Using these architectures may yield better results.
- **Hyperparameter Optimization:** The dropout rate, learning rate, and number of neurons per layer were manually chosen. Using grid search or Bayesian optimization could find more optimal configurations.
- **Additional Features:** Combining MFCCs and Mel-spectrograms with other features such as pitch, energy, and formants might offer richer representations for classification.

By exploring these avenues, future models could potentially surpass the performance of the current baseline and offer more robust speech gender recognition systems suitable for deployment in real-world scenarios.